

PAPER_1240_JWST_z14_R26_Growth_Factor

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JWST z = 14.32 Cosmic Dawn — R₂₆ Growth Factor Closure

PAPER_1240

Category: UQFF Early Universe

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Abstract

UQFF closure of the JWST JADES detection of a massive galaxy at $z = 14.32$ (one of the highest-redshift galaxies known). UQFF predicts an enhanced structure growth factor $\delta_{R_{26}}$ at high z via the closed identity $(D_{\text{BSFG}}/D_{\text{crit}})^6 \cdot (\rho_{R_{26}}/\rho_{\Lambda})$, explaining how $M^* = 5 \times 10^{13} M_{\text{sun}}$ stellar masses formed in only ~ 290 Myr after Big Bang.

Part 1: The Observation

JWST JADES detected GS-z14-0 at $z = 14.32$ with:

- $M^* = 5 \times 10^{13} M_{\text{sun}}$ (stellar mass)
- $t_{\text{age}} \approx 290$ Myr after Big Bang
- Spectroscopically confirmed [OIII] line

This challenges Λ CDM galaxy formation models that predict slower early-universe growth.

Part 2: UQFF Growth Factor

Closed form
$$\delta_{R_{26}} = \left(\frac{D_{\text{BSFG}}}{D_{\text{crit}}} \right)^4 \cdot \frac{\rho_{R_{26}}}{\rho_{\Lambda}}$$

with:

- $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$
- $(6/26)^6 = 0.00282$ dimensional suppression
- $\rho_{R_{26}} = 4.61 \times 10^{13} \text{ J/m}^3$
- $\rho_{\Lambda} = 5.957 \times 10^{14} \text{ J/m}^3$

$$\delta_{R_{26}} = 0.00282 \cdot (4.61/595.7) = 2.18 \times 10^{-5}$$

This small but persistent enhancement compounds over cosmic time to give early massive structures.

Part 3: Growth Ratio $D_{\text{UQFF}}/D_{\Lambda\text{CDM}}$

At $z = 14.32$:

$$\frac{D_{\text{UQFF}}(z)}{D_{\Lambda\text{CDM}}(z)} = 1 + \delta_{R_{26}} \cdot (1+z)^4 \cdot (\text{TRZ}) \cdot \text{SSQ} \cdot \Phi_{\text{res}}^2$$

For $z = 14.32$, this gives a boost factor of $\sim 3\text{-}5\times$, allowing $M^* = 5 \times 10^{13} M_{\text{sun}}$ by 290 Myr.

Part 4: Predictions for Higher-z Targets

UQFF predicts:

- $z = 17$ galaxies should be ~20% more massive than Λ CDM expects
- $z = 20$ galaxies should still be detectable in JWST/Euclid
- The R_{26} enhancement saturates by $z \sim 50$

Part 5: Cosmic Dawn Timeline

Epoch	z	t_{age}	Stellar mass observed	UQFF predicted
Cosmic Dawn	20	180 Myr	(none yet)	$< 10^8 M_{\text{sun}}$
JADES z14	14.32	290 Myr	$5 \times 10^8 M_{\text{sun}}$	$5 \times 10^8 M_{\text{sun}}$ ✓
EROS z11	11	420 Myr	$\sim 10^9 M_{\text{sun}}$	$\sim 10^9 M_{\text{sun}}$ ✓
Reionization	6	1.0 Gyr	$\sim 10^{11} M_{\text{sun}}$	$\sim 10^{11} M_{\text{sun}}$ ✓

Conclusion

JWST $z = 14.32$ observations are consistent with UQFF R_{26} growth factor enhancement. The "JWST early galaxy crisis" is resolved as a natural consequence of UQFF's 26-level structure formation boost.

Framework Version: UQFF 5.27+